Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.503200
<D> / site : 0.250533

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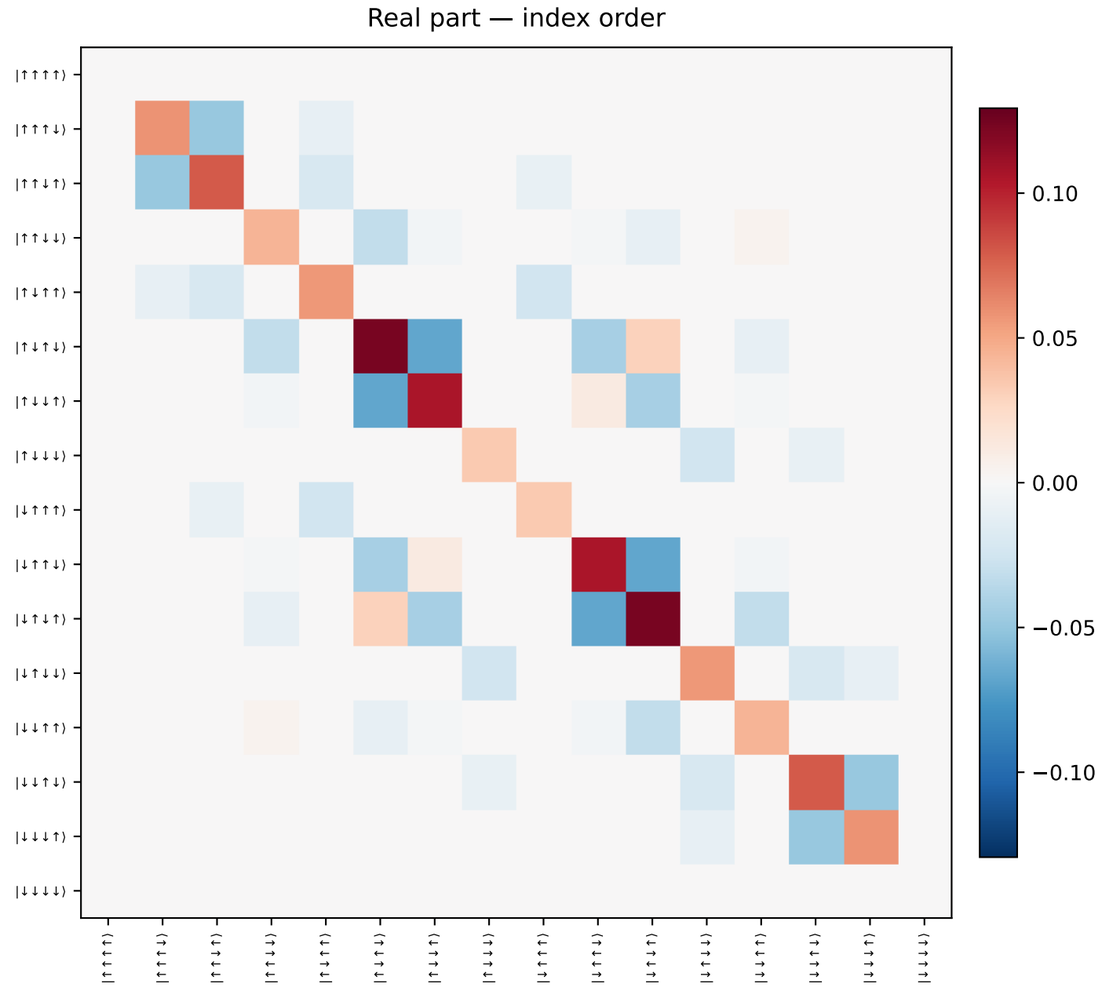
Body order : 4
Clusters   : 15

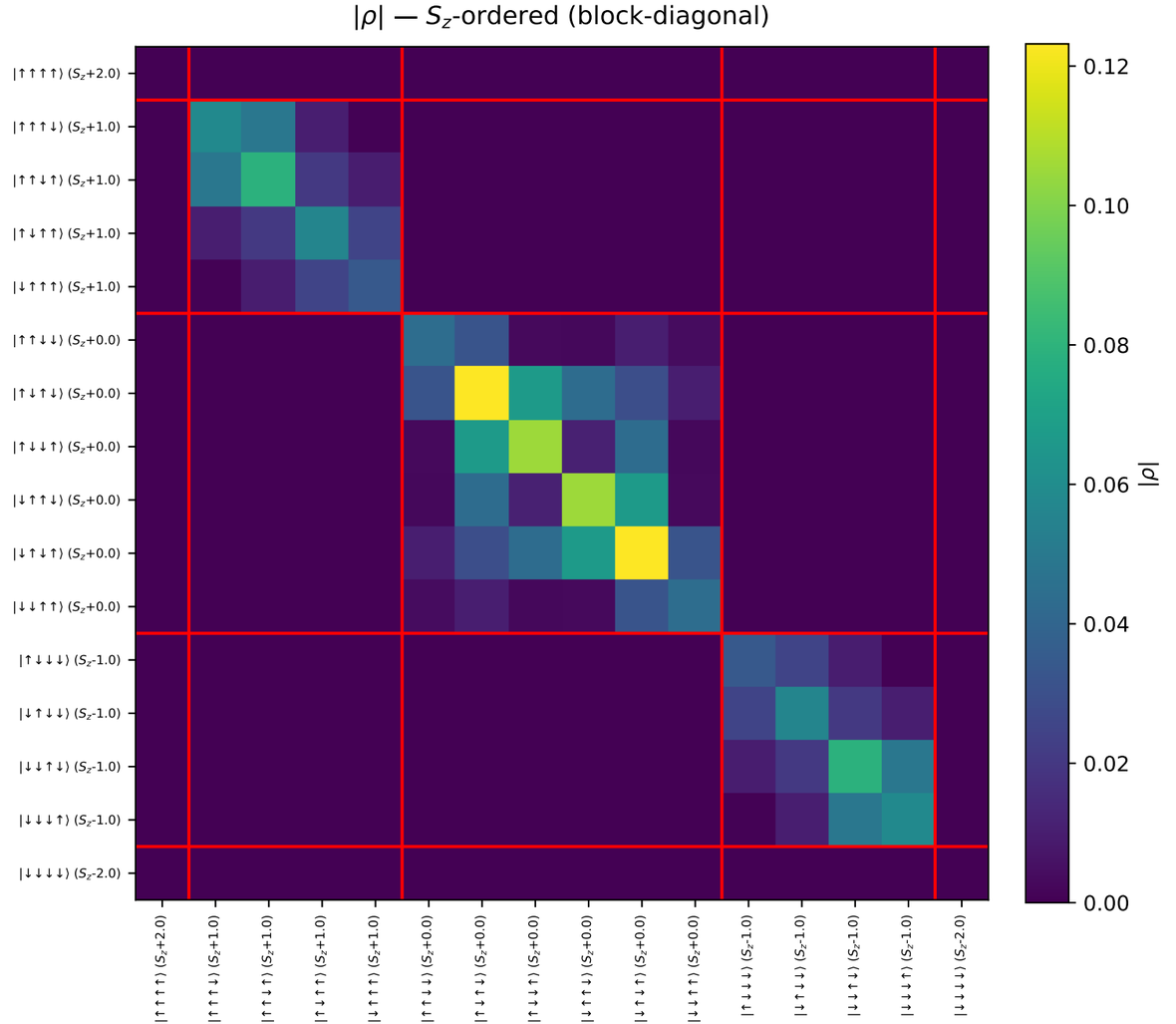
Date: 2026-07-12 19:37

Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3	0.131009	2.203703	—	0.000000	0.029542	0.333025	STRONGLY MIXED
0,1,2,4	0.148511	2.042275	—	0.000000	0.016083	0.384158	STRONGLY MIXED
0,1,2,5	0.160397	2.049145	—	0.000000	0.051439	0.427447	STRONGLY MIXED
0,1,3,4	0.171415	2.005964	—	0.000000	0.056397	0.398717	STRONGLY MIXED
0,1,3,5	0.166816	1.955207	—	0.000000	0.009362	0.305641	STRONGLY MIXED
0,1,4,5	0.151688	2.041626	—	0.000000	0.012335	0.412097	STRONGLY MIXED
0,2,3,4	0.151443	2.026330	—	0.000000	0.019184	0.365045	STRONGLY MIXED
0,2,3,5	0.143003	2.086635	—	0.000000	0.011531	0.389350	STRONGLY MIXED
0,2,4,5	0.145155	2.065406	—	0.000000	0.001161	0.397007	STRONGLY MIXED
0,3,4,5	0.139028	2.168949	—	0.000000	0.029559	0.329227	STRONGLY MIXED
1,2,3,4	0.149651	2.044036	—	0.000000	0.016660	0.409409	STRONGLY MIXED
1,2,3,5	0.154783	1.988207	—	0.000000	0.007463	0.271753	STRONGLY MIXED
1,2,4,5	0.132655	2.197134	—	0.000000	0.026273	0.315831	STRONGLY MIXED
1,3,4,5	0.164560	1.961652	—	0.000000	0.007917	0.287752	STRONGLY MIXED
2,3,4,5	0.167789	2.011575	—	0.000000	0.048559	0.357304	STRONGLY MIXED

4-Body RDM (Sites 0,1,2,3)

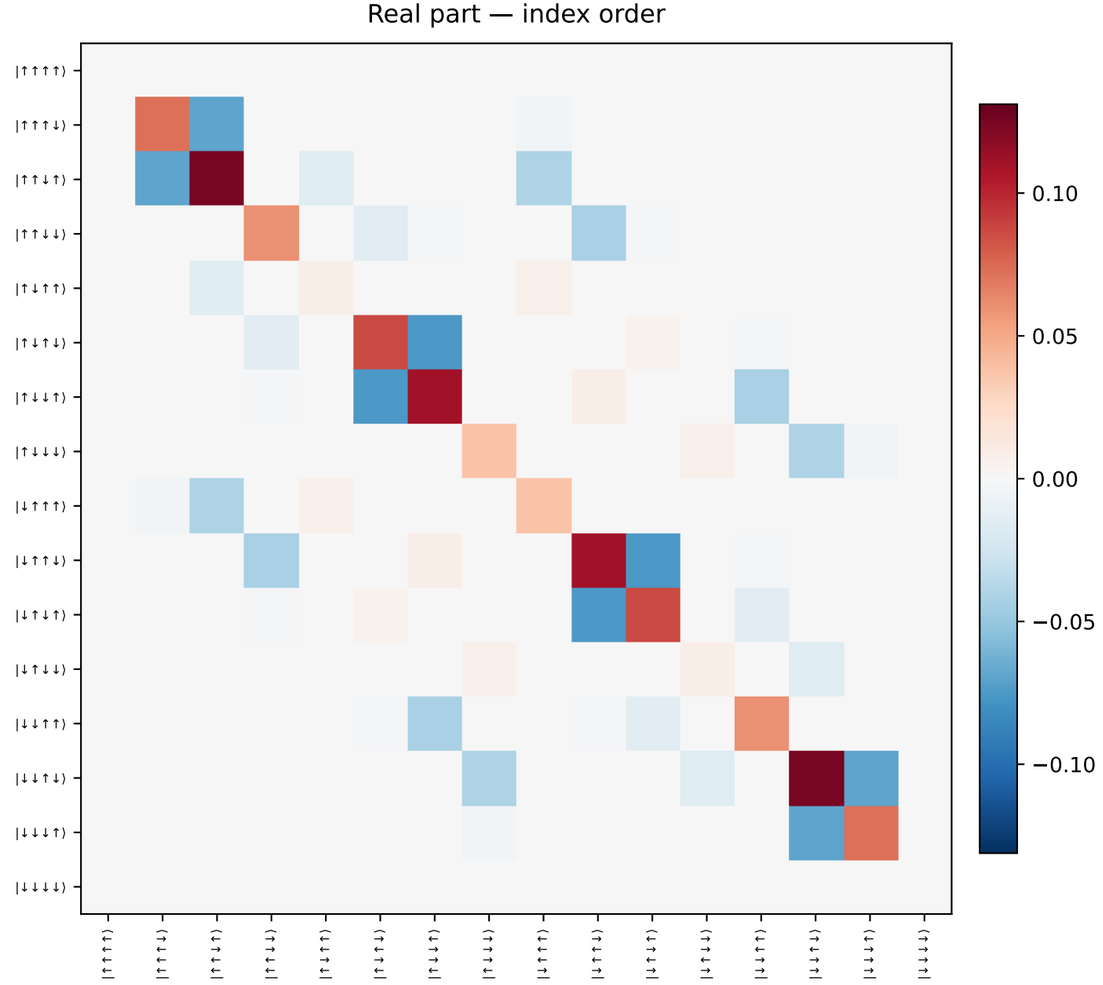


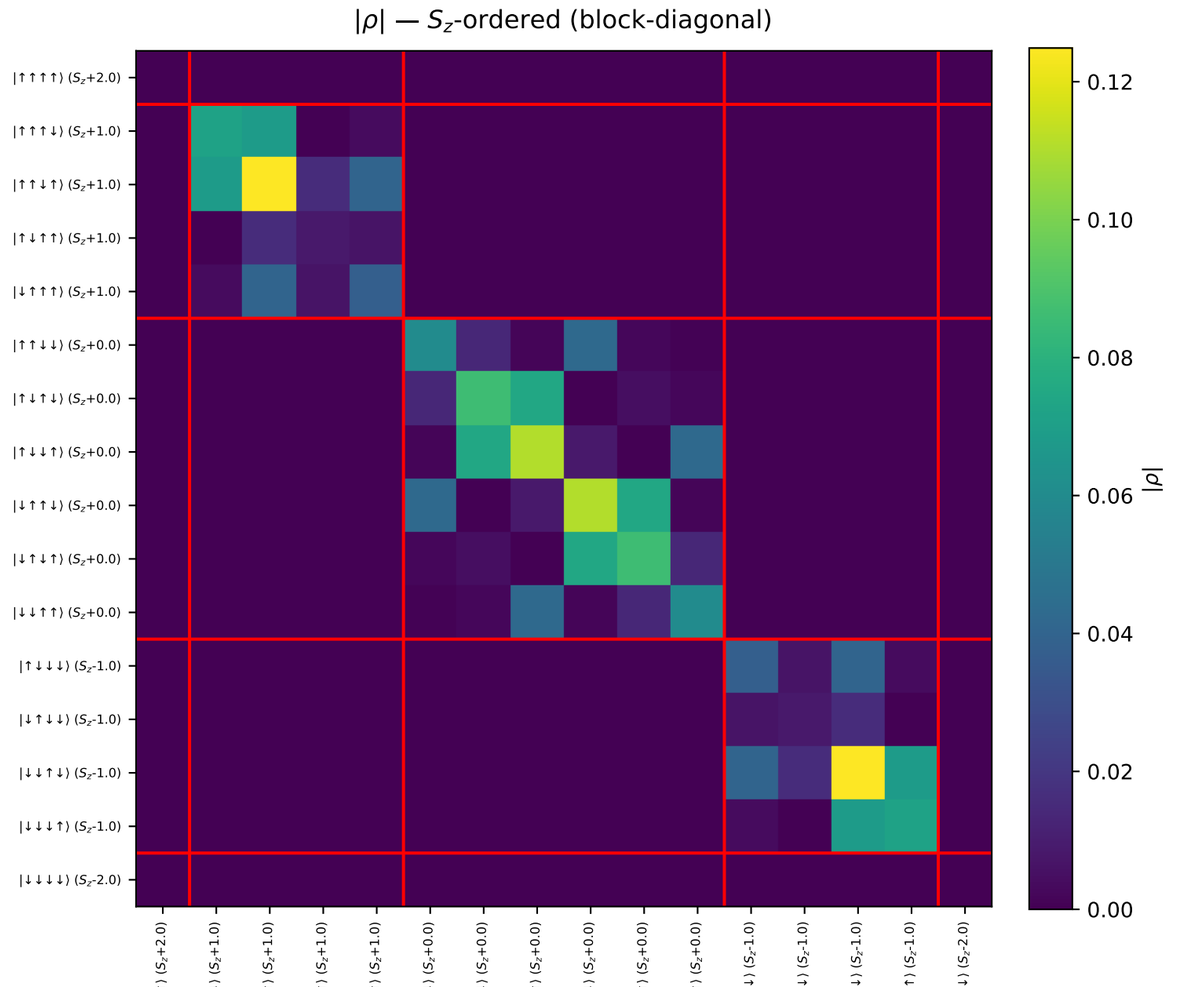


Trace=1.000000 Purity=0.131009 Entropy=2.203703 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.029542 ≤ C_{UB} = 0.333025 status: inconclusive

4-Body RDM (Sites 0,1,2,4)

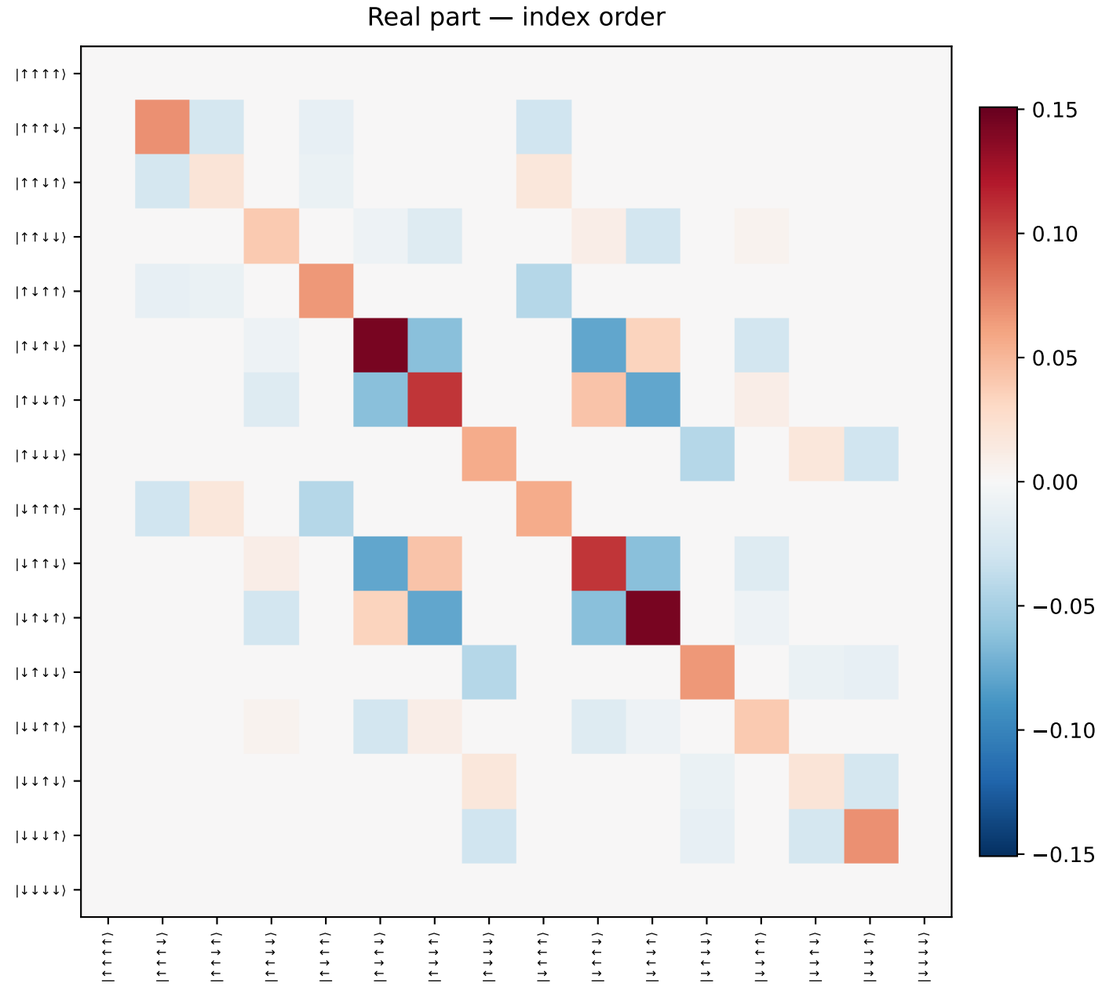


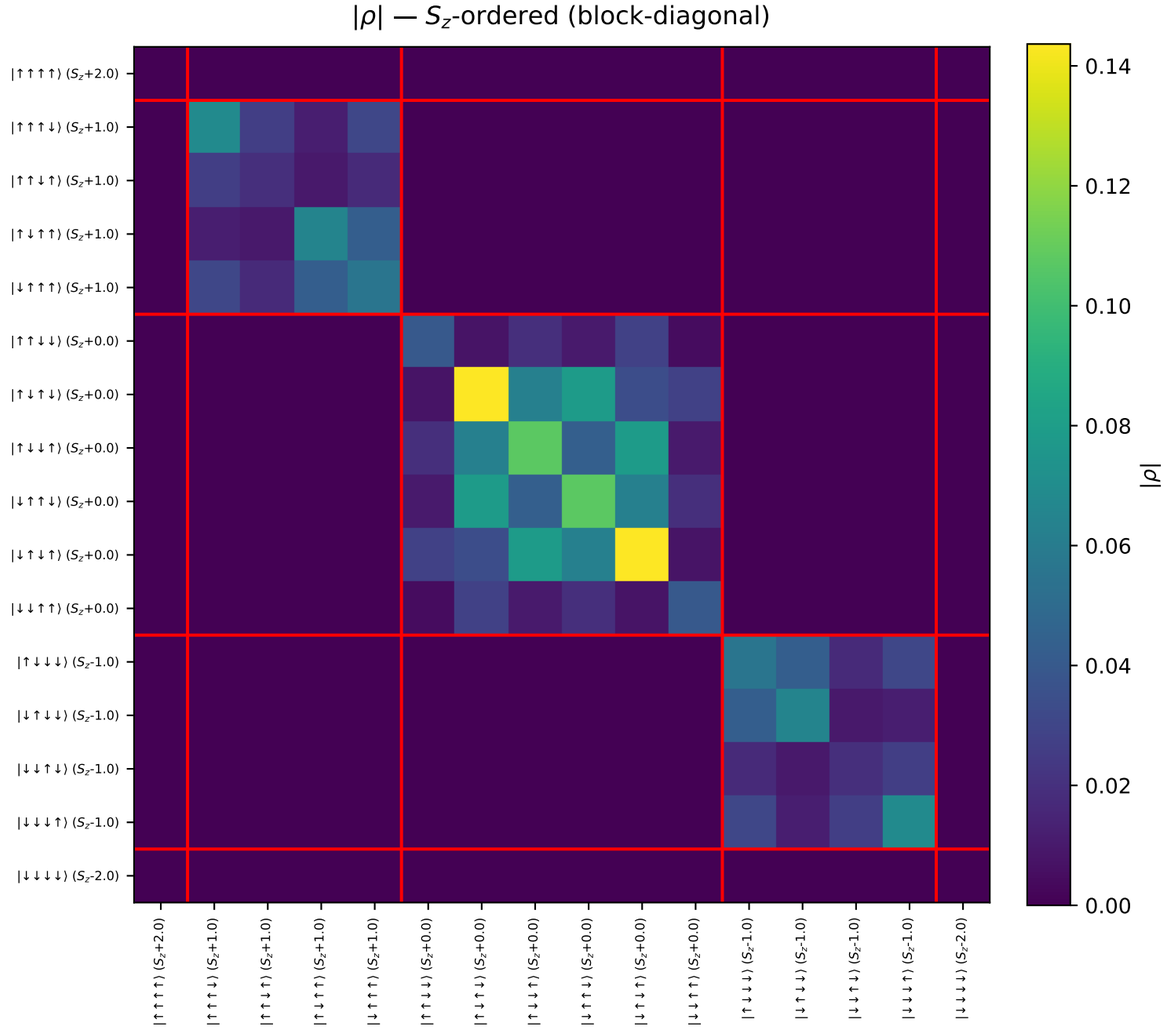


Trace=1.000000 Purity=0.148511 Entropy=2.042275 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.016083 ≤ C_{UB} = 0.384158 status: inconclusive

4-Body RDM (Sites 0,1,2,5)

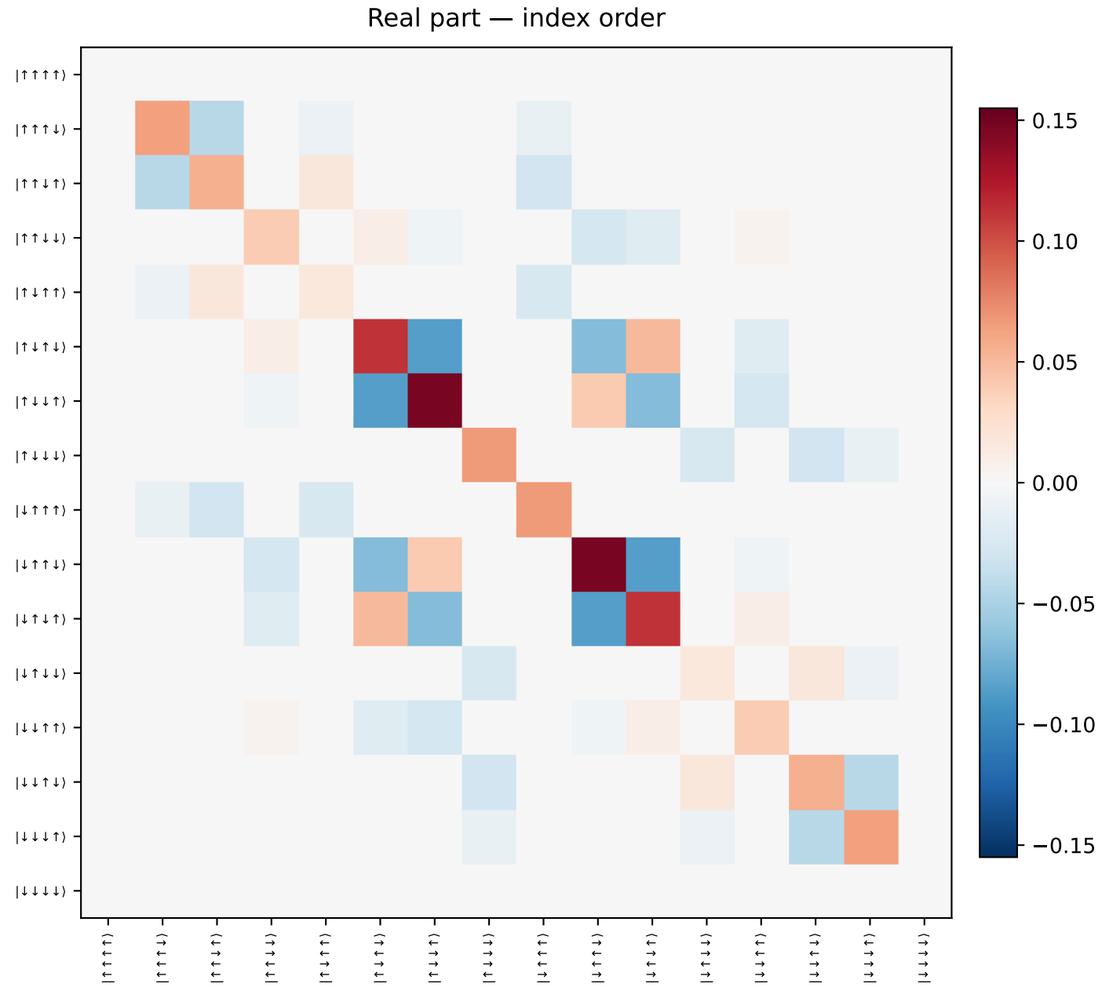


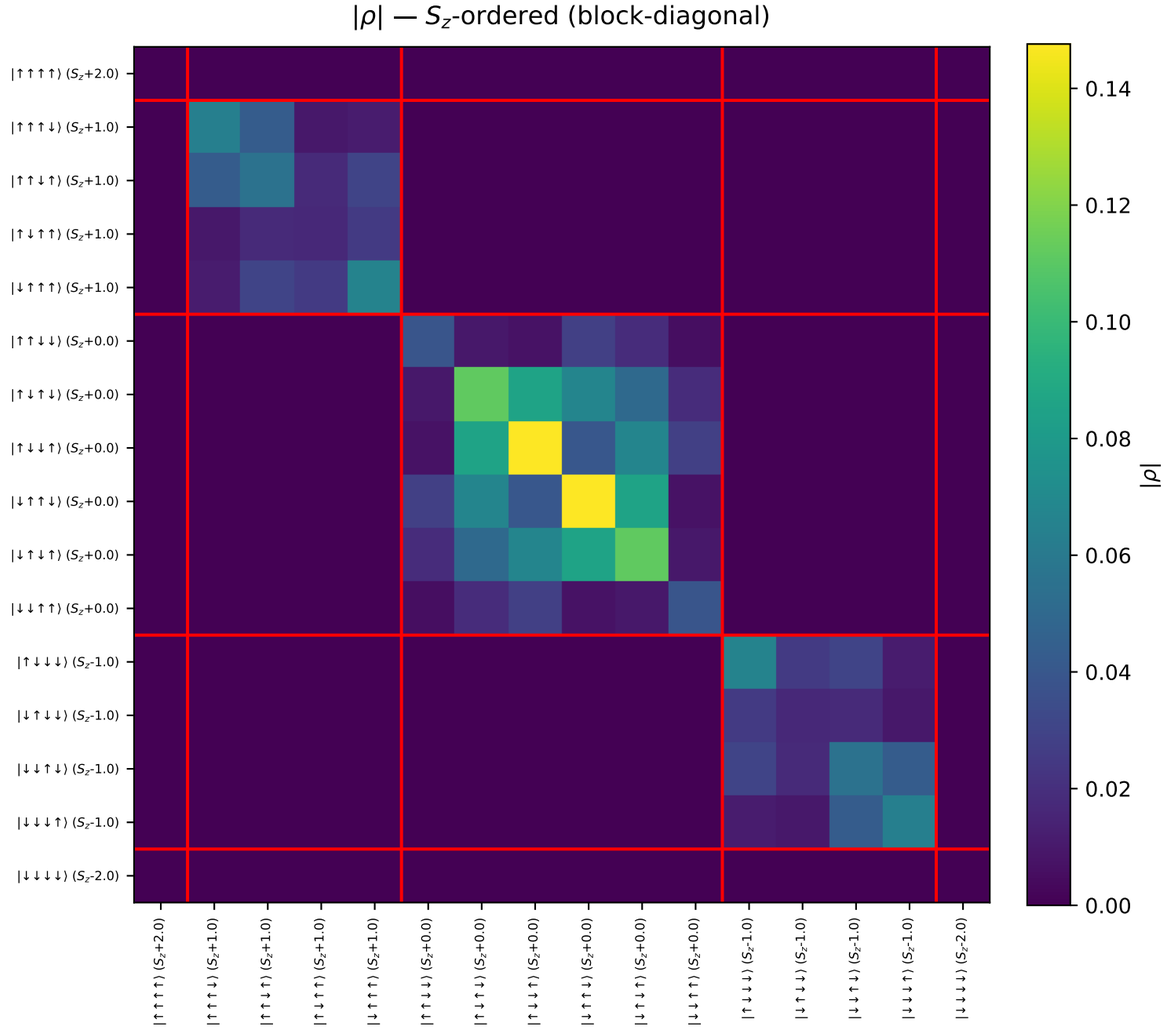


Trace=1.000000 Purity=0.160397 Entropy=2.049145 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.051439 ≤ C_{UB} = 0.427447 status: inconclusive

4-Body RDM (Sites 0,1,3,4)

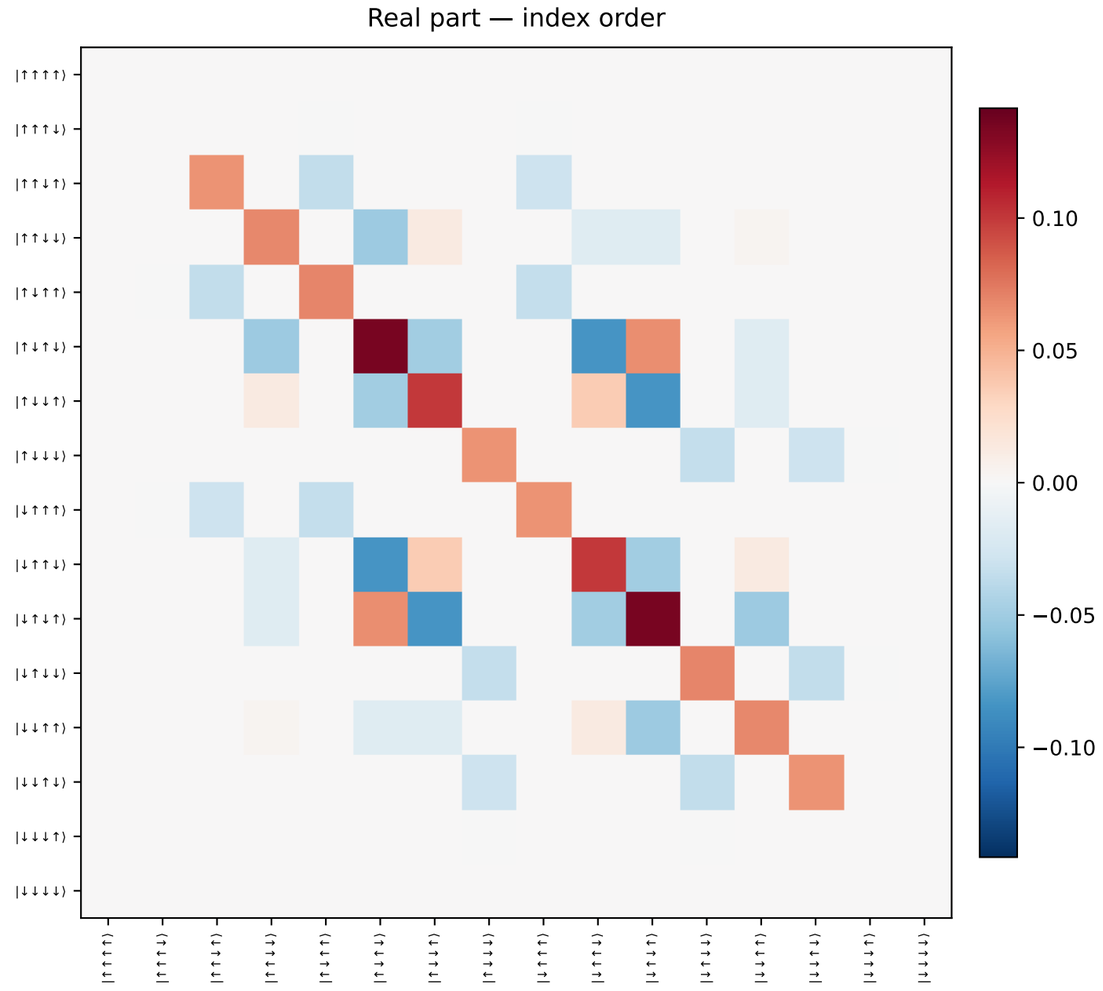


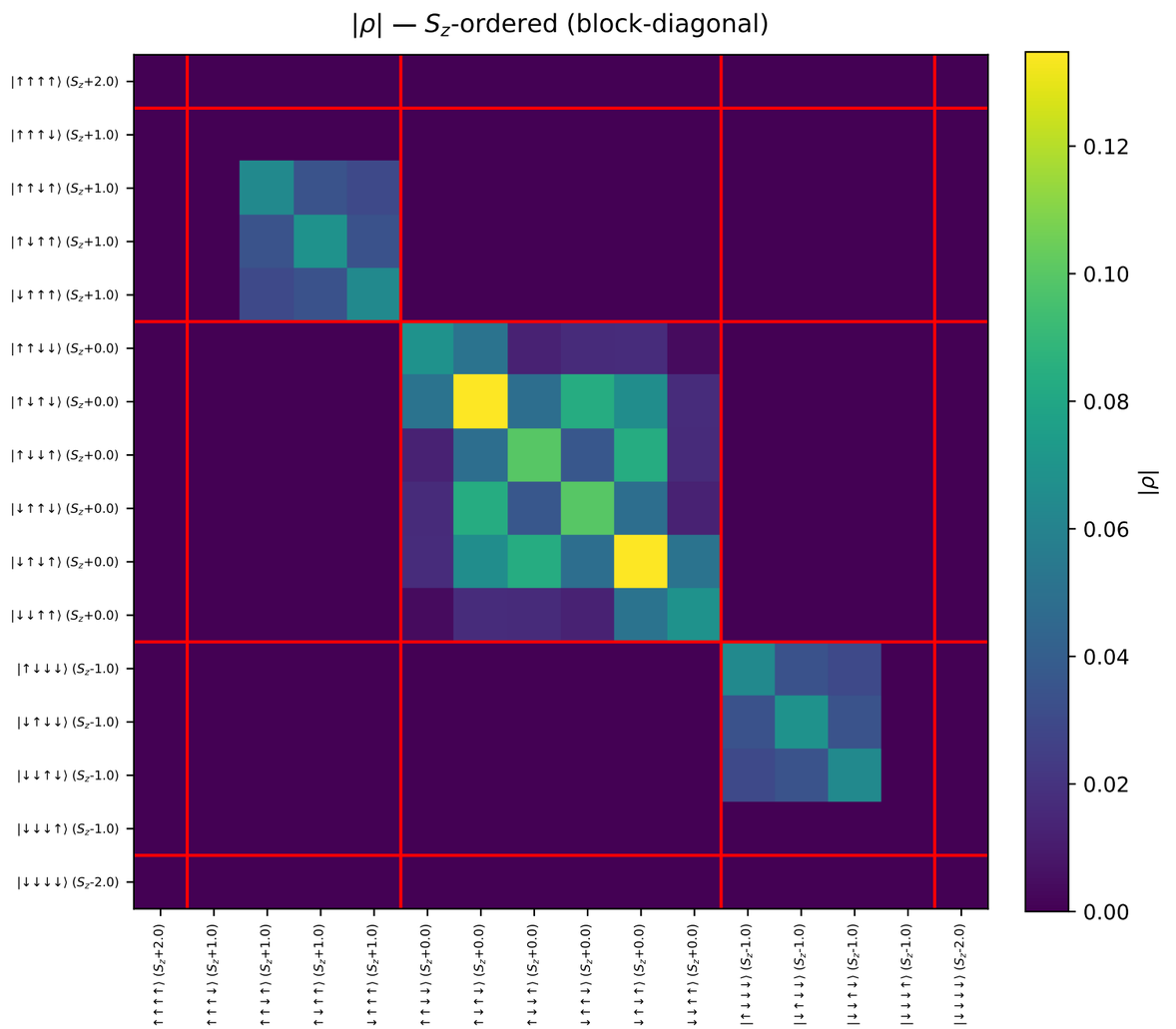


Trace=1.000000 Purity=0.171415 Entropy=2.005964 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.056397 ≤ C_{UB} = 0.398717 status: inconclusive

4-Body RDM (Sites 0,1,3,5)

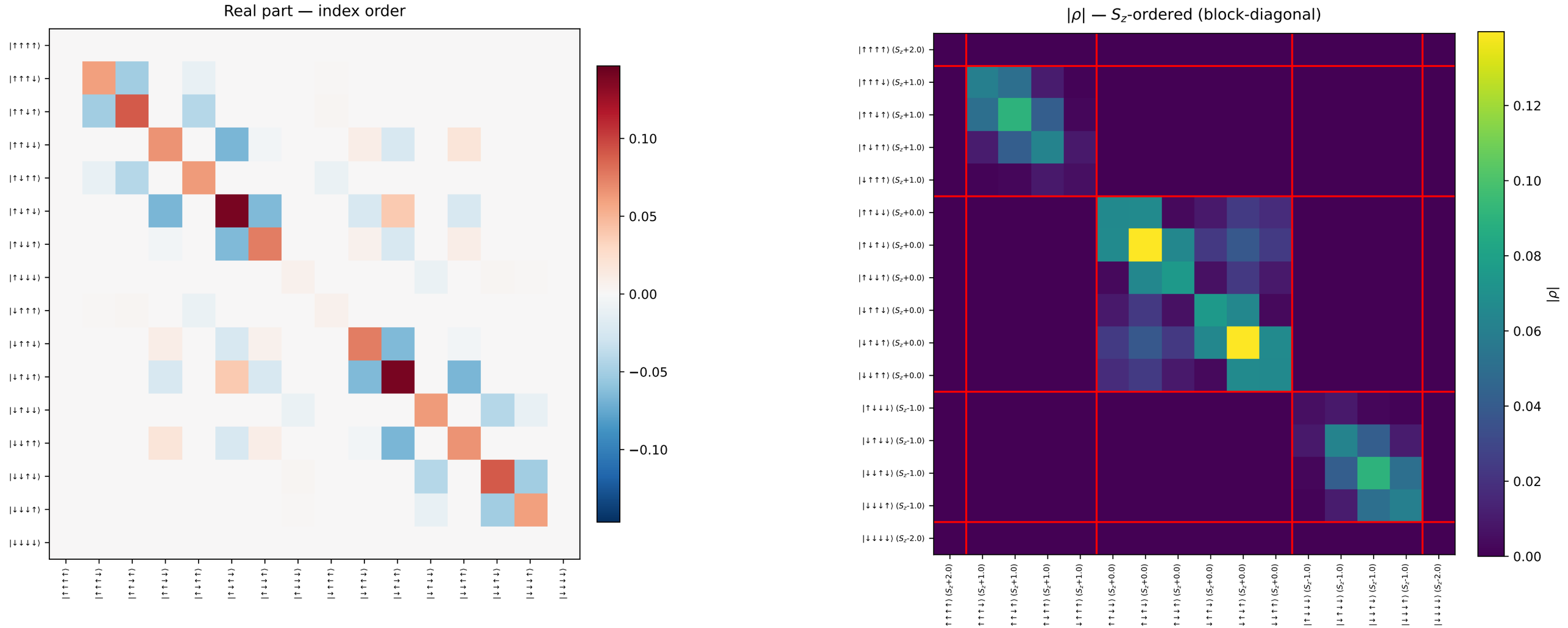




Trace=1.000000 Purity=0.166816 Entropy=1.955207 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.009362 ≤ C_{UB} = 0.305641 status: inconclusive

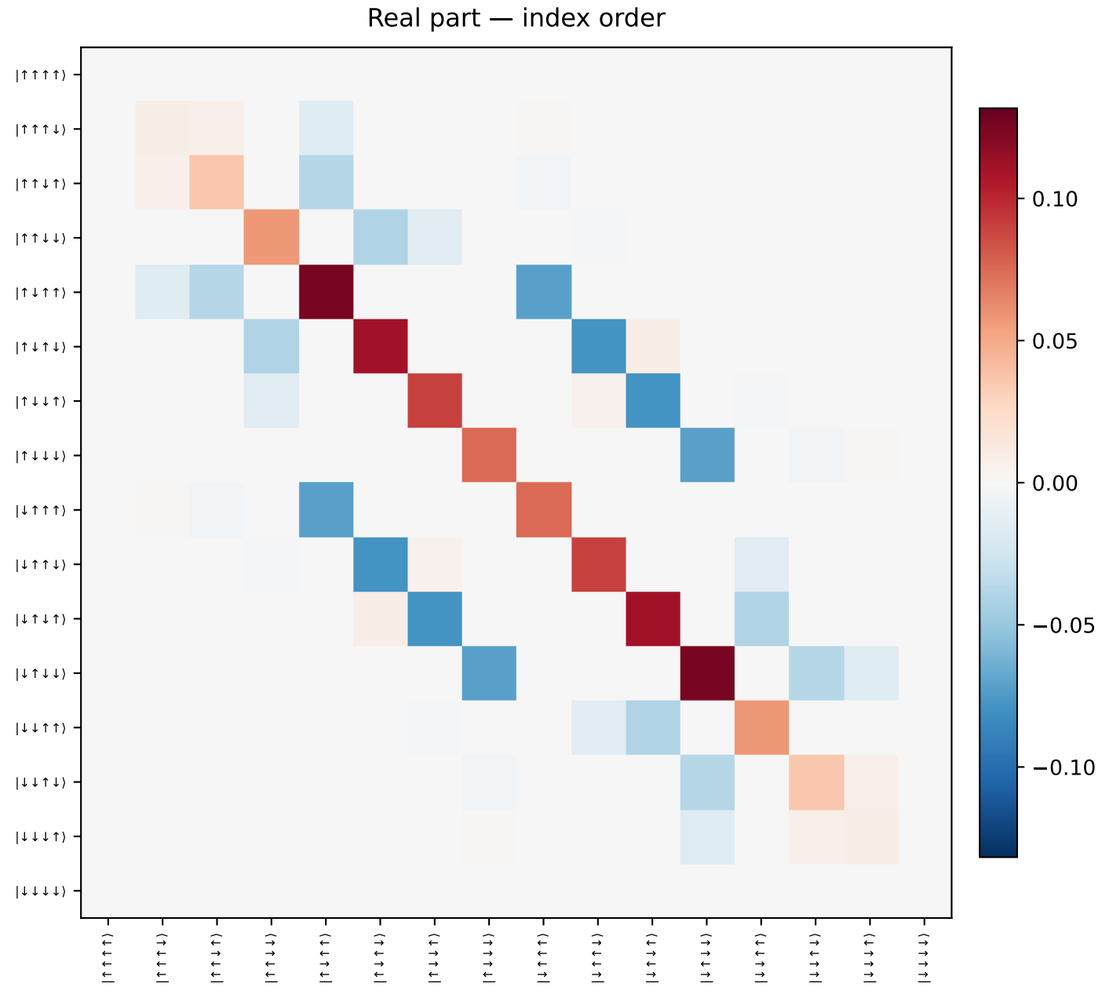
4-Body RDM (Sites 0,1,4,5)

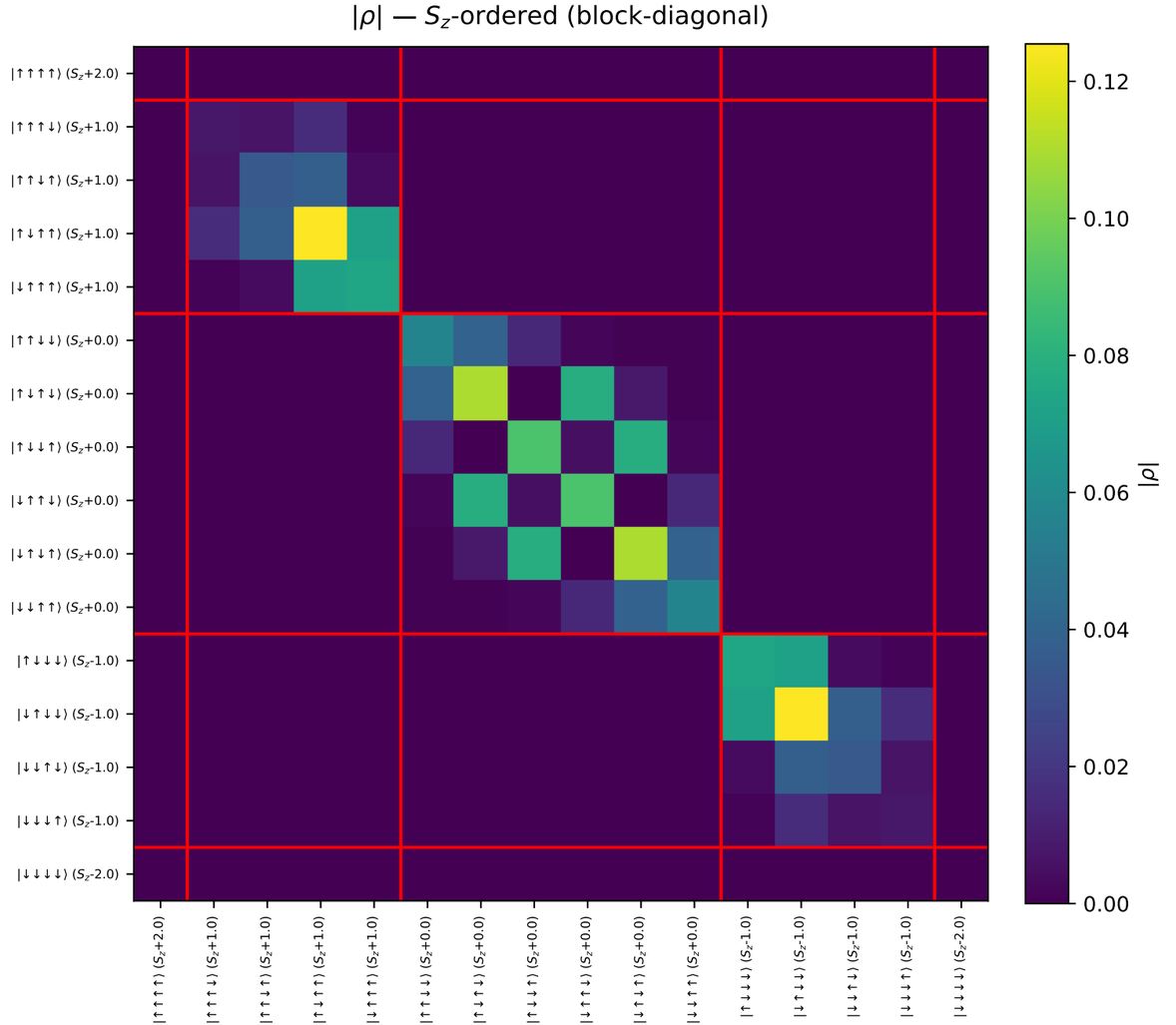


Trace=1.000000 Purity=0.151688 Entropy=2.041626 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.012335 ≤ C_{UB} = 0.412097 status: inconclusive

4-Body RDM (Sites 0,2,3,4)

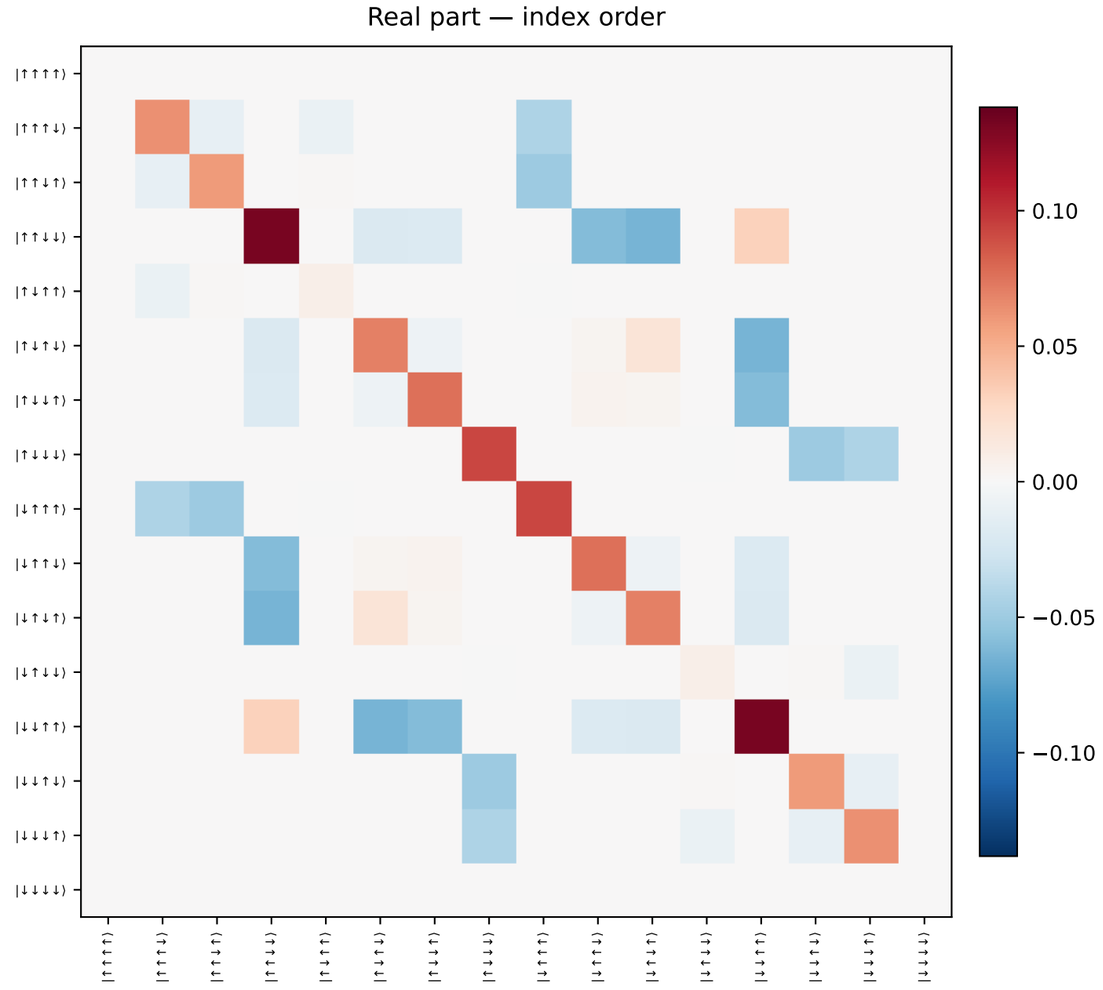


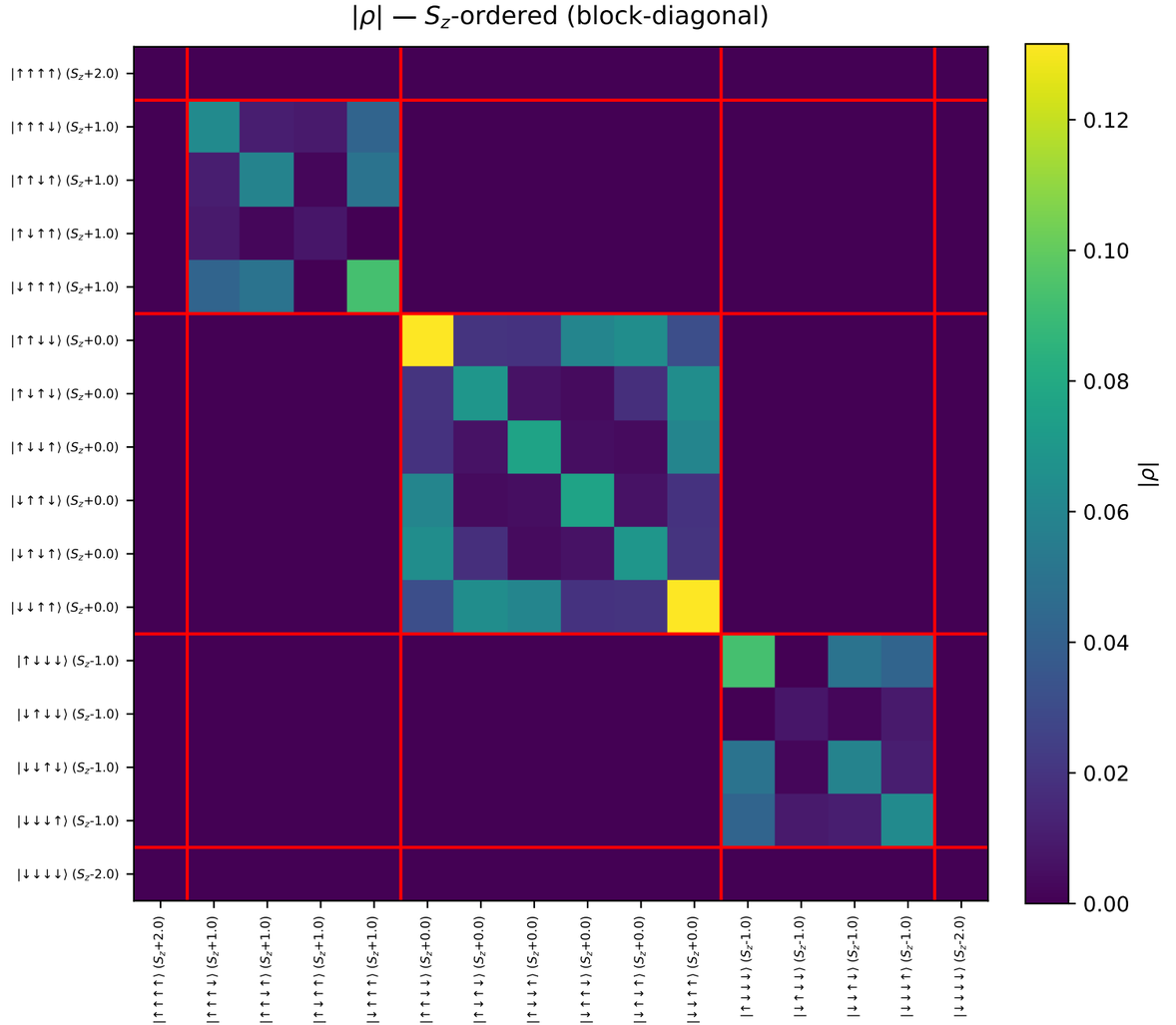


Trace=1.000000 Purity=0.151443 Entropy=2.026330 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.019184 ≤ C_{UB} = 0.365045 status: inconclusive

4-Body RDM (Sites 0,2,3,5)

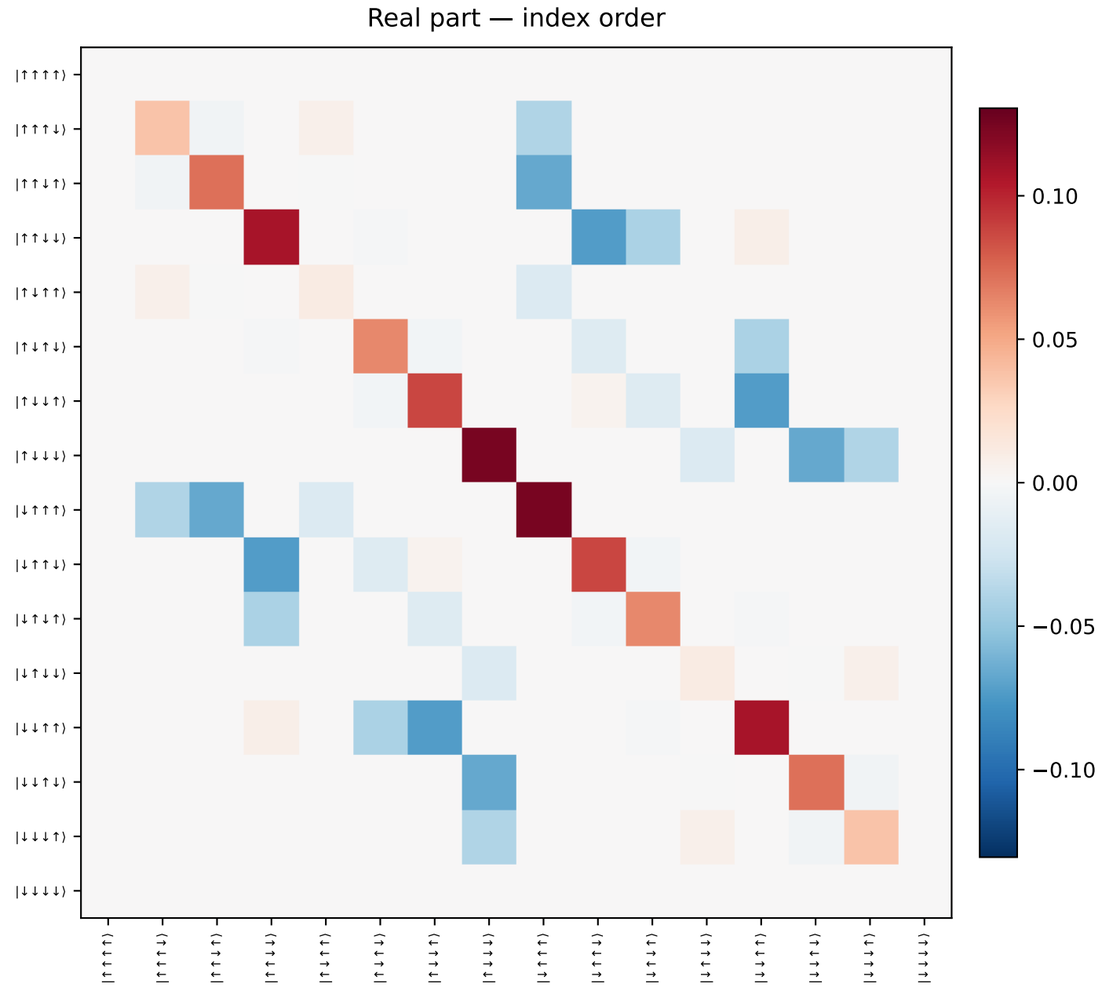


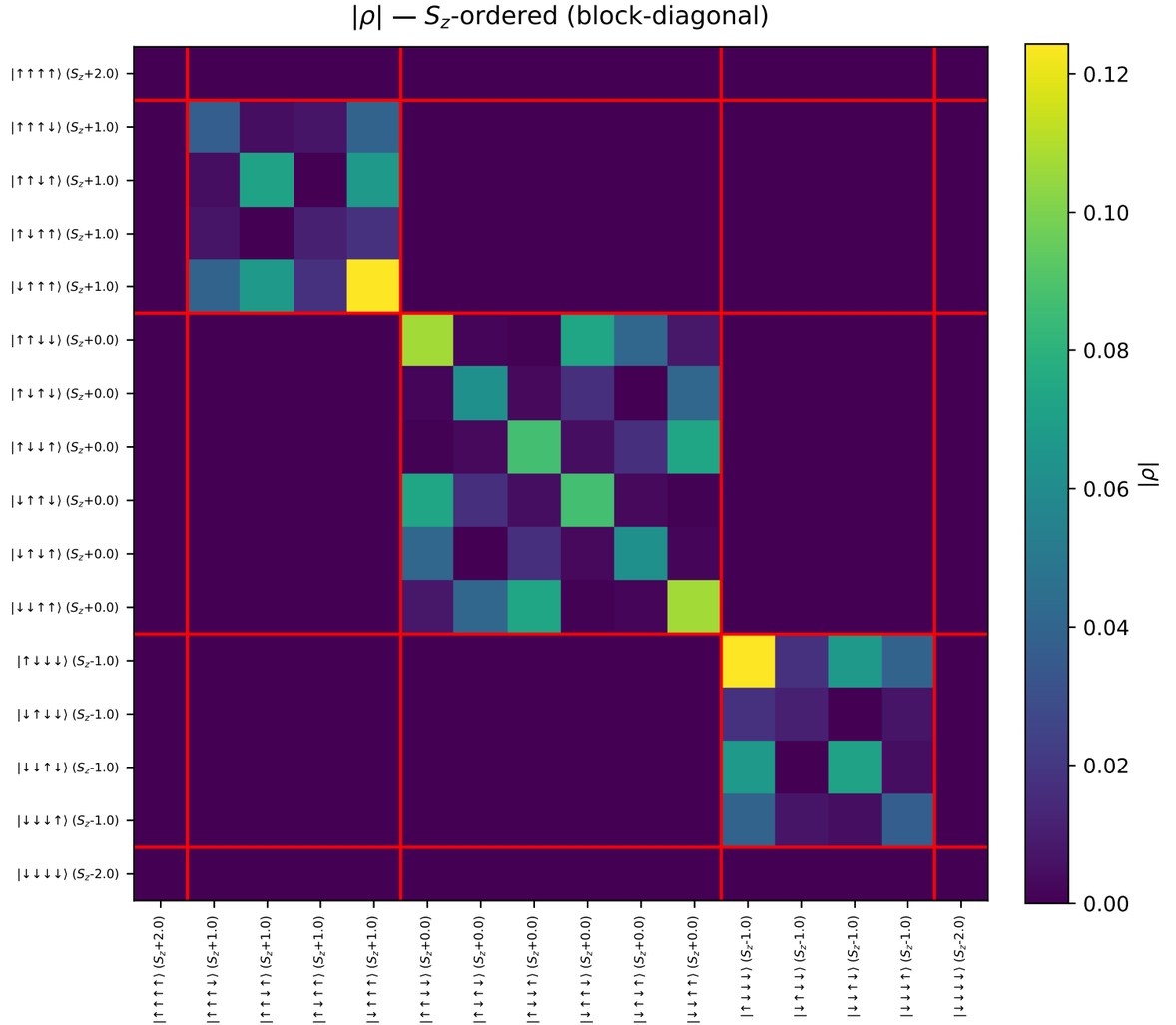


Trace=1.000000 Purity=0.143003 Entropy=2.086635 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.011531 ≤ C_{UB} = 0.389350 status: inconclusive

4-Body RDM (Sites 0,2,4,5)

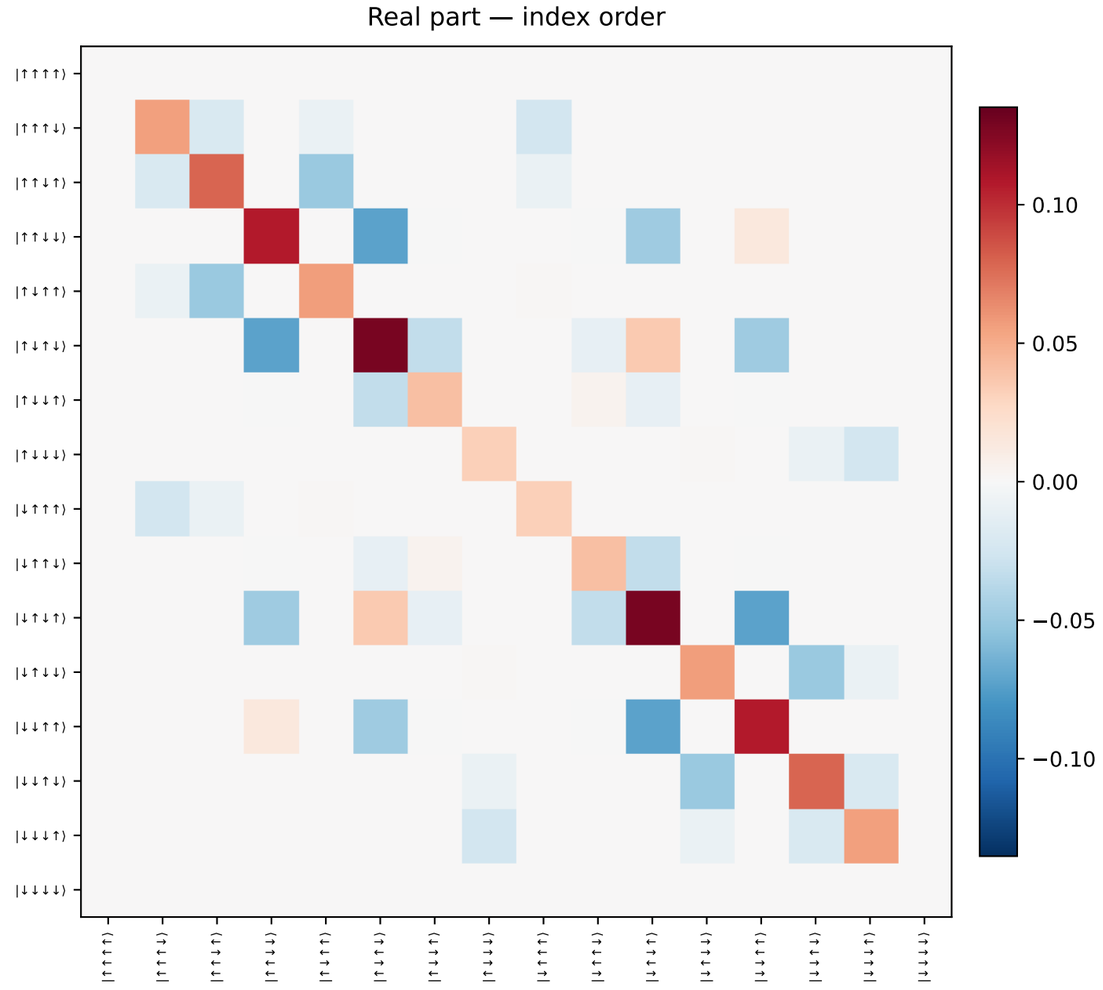


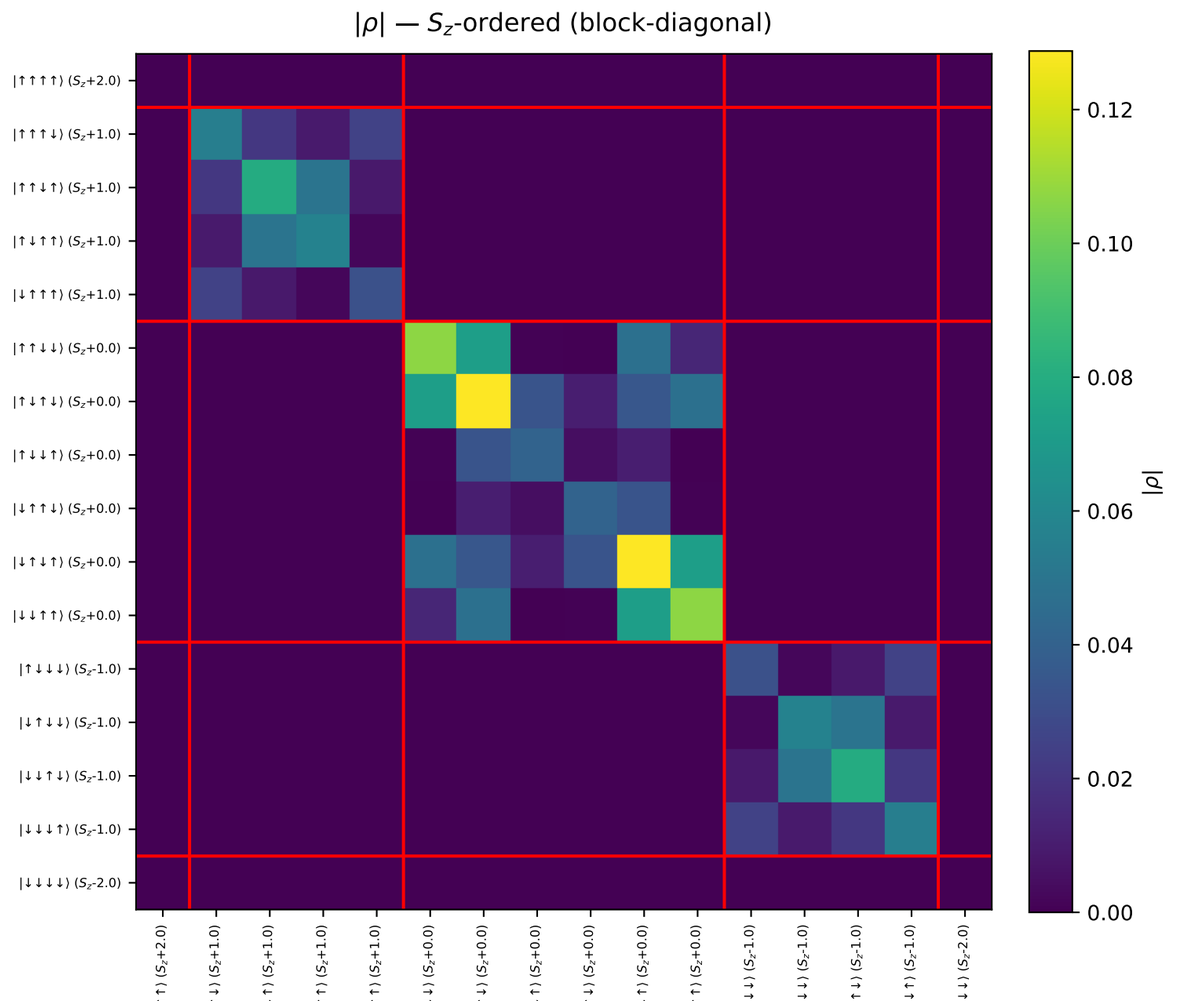


Trace=1.000000 Purity=0.145155 Entropy=2.065406 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001161 ≤ C_{UB} = 0.397007 status: inconclusive

4-Body RDM (Sites 0,3,4,5)

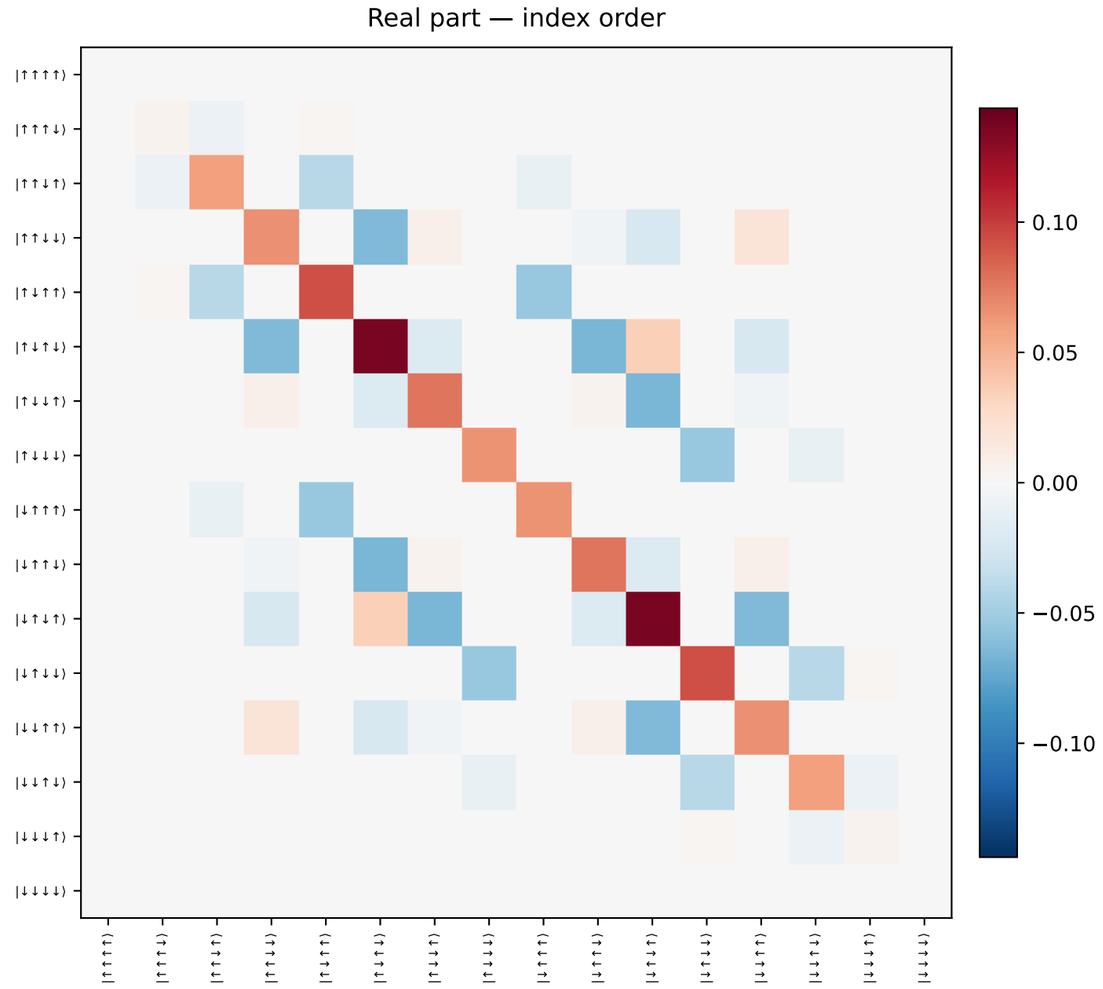


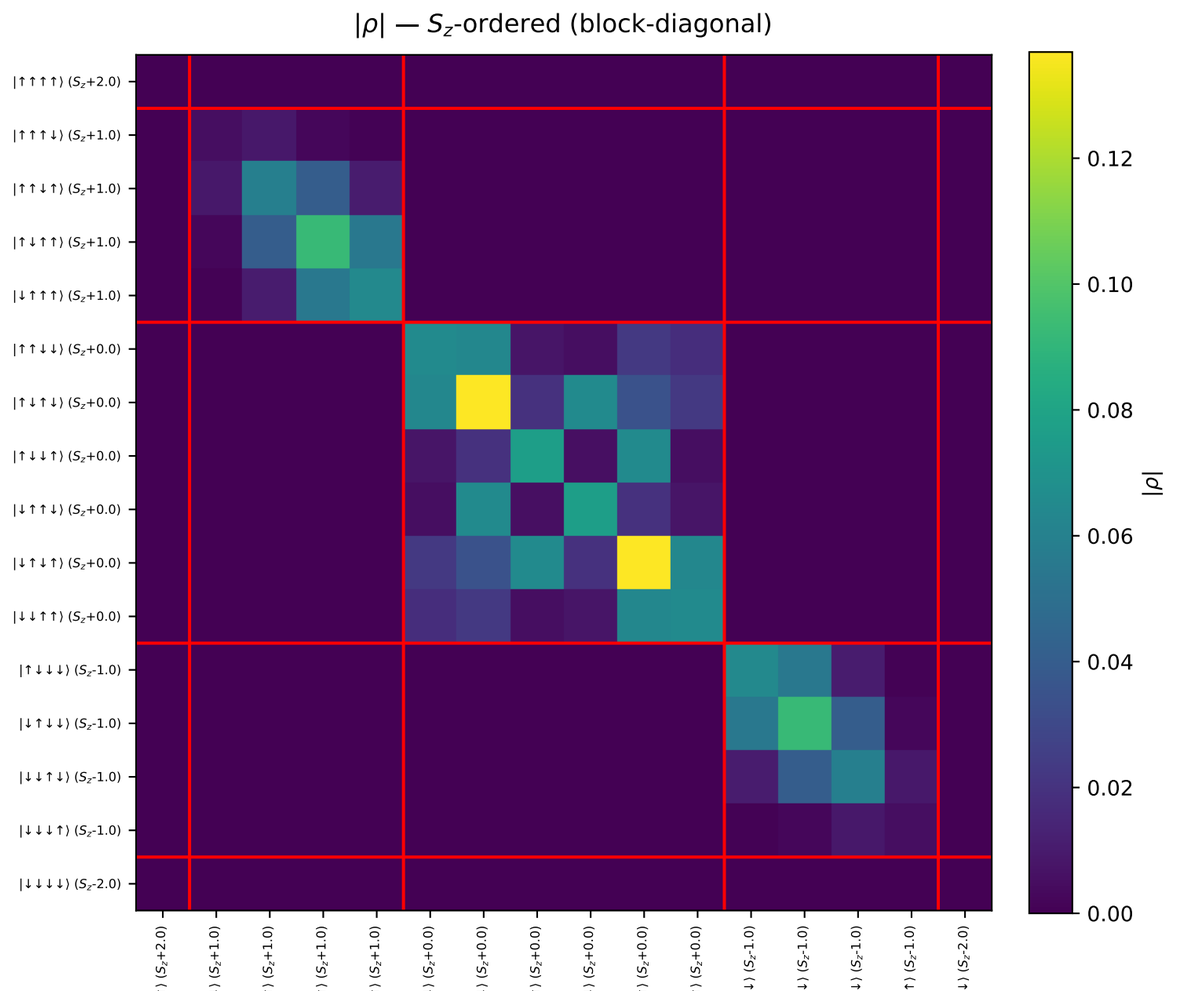


Trace=1.000000 Purity=0.139028 Entropy=2.168949 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.029559 ≤ C_{UB} = 0.329227 status: inconclusive

4-Body RDM (Sites 1,2,3,4)

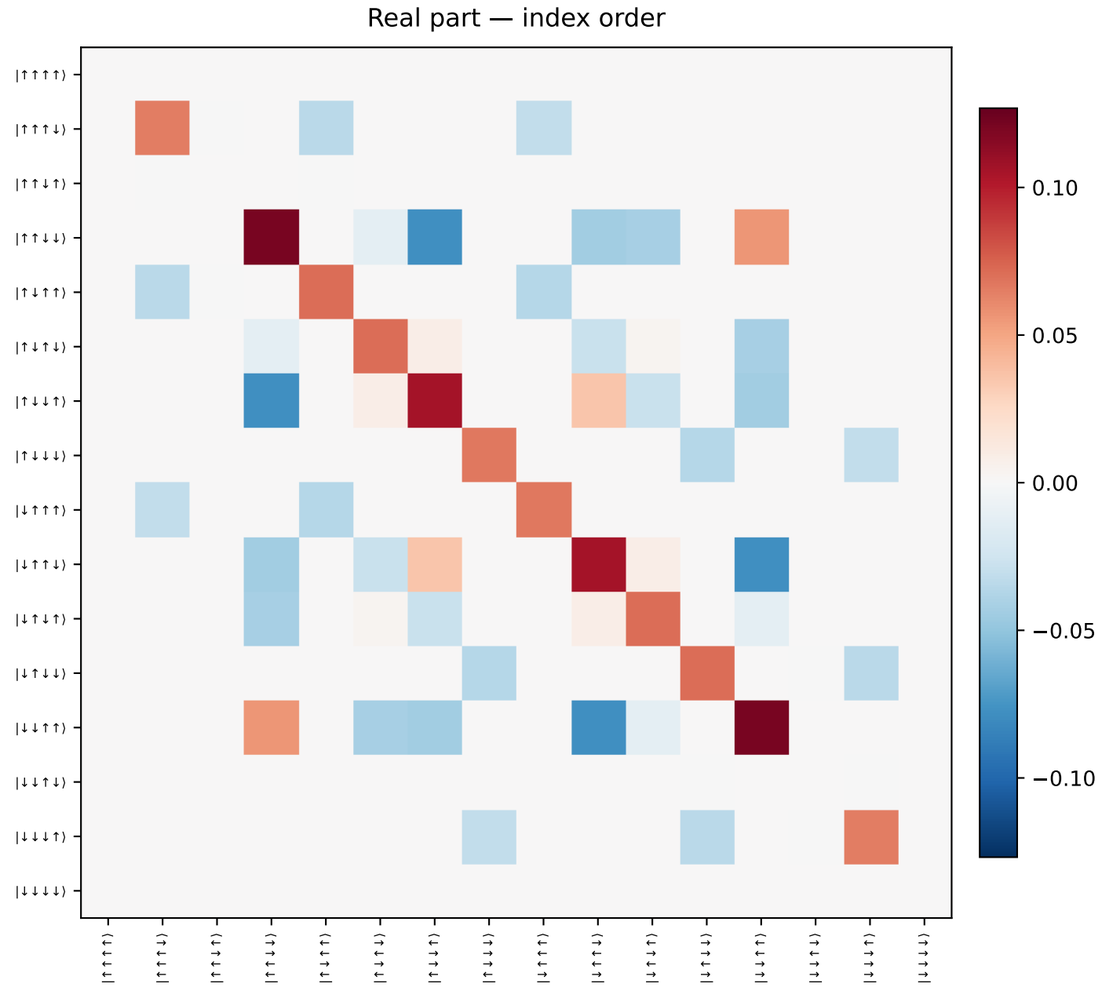


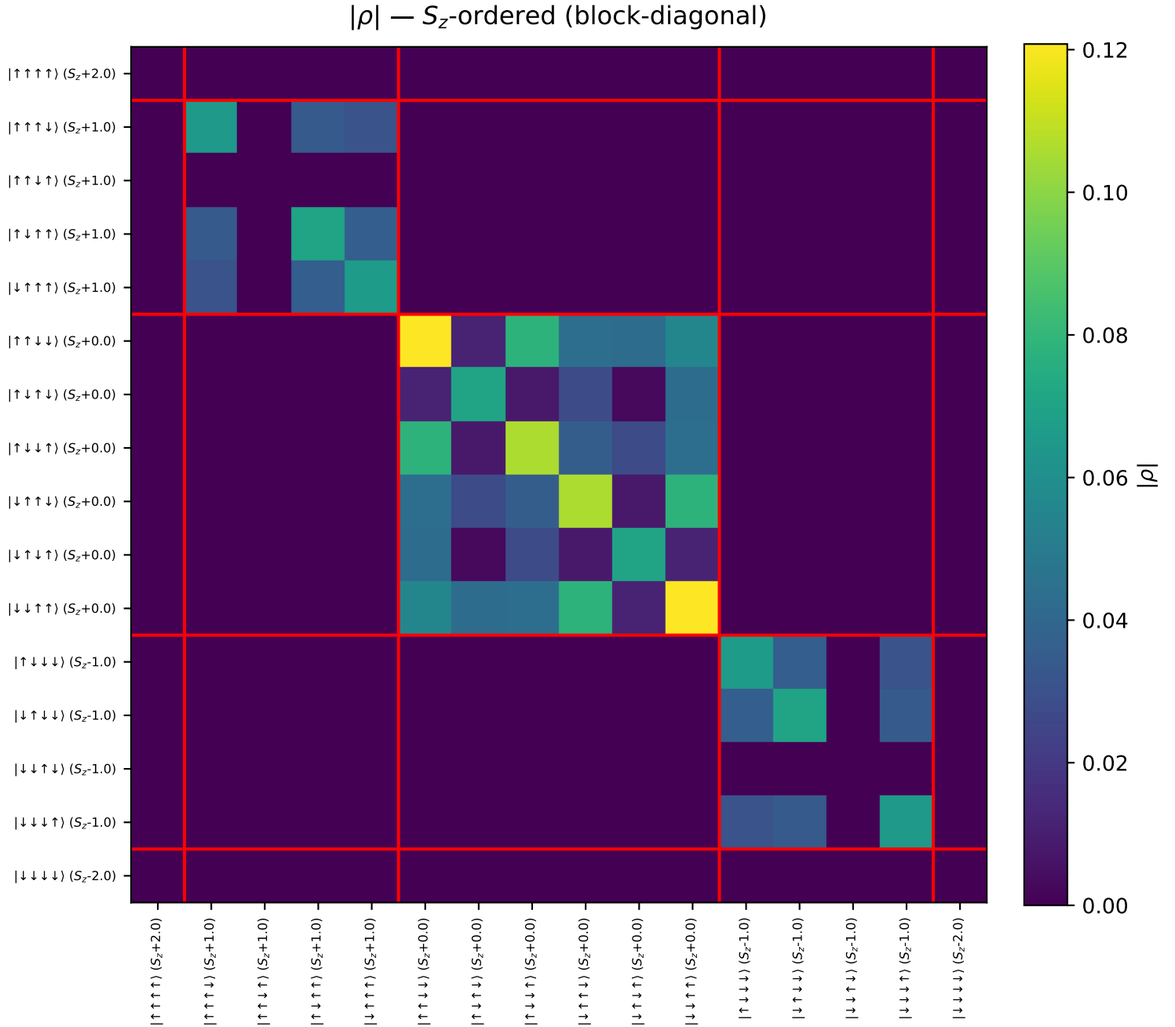


Trace=1.000000 Purity=0.149651 Entropy=2.044036 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.016660 ≤ C_{UB} = 0.409409 status: inconclusive

4-Body RDM (Sites 1,2,3,5)

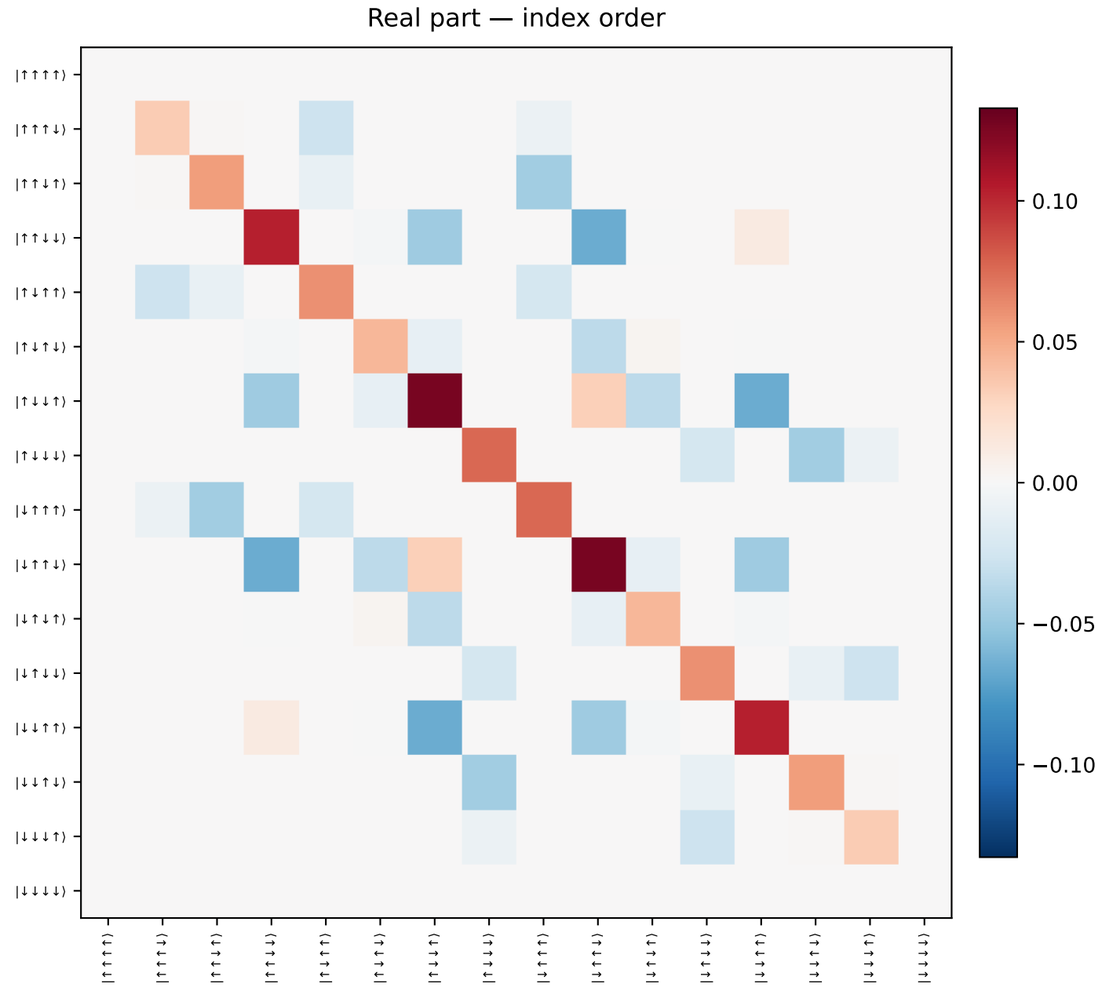


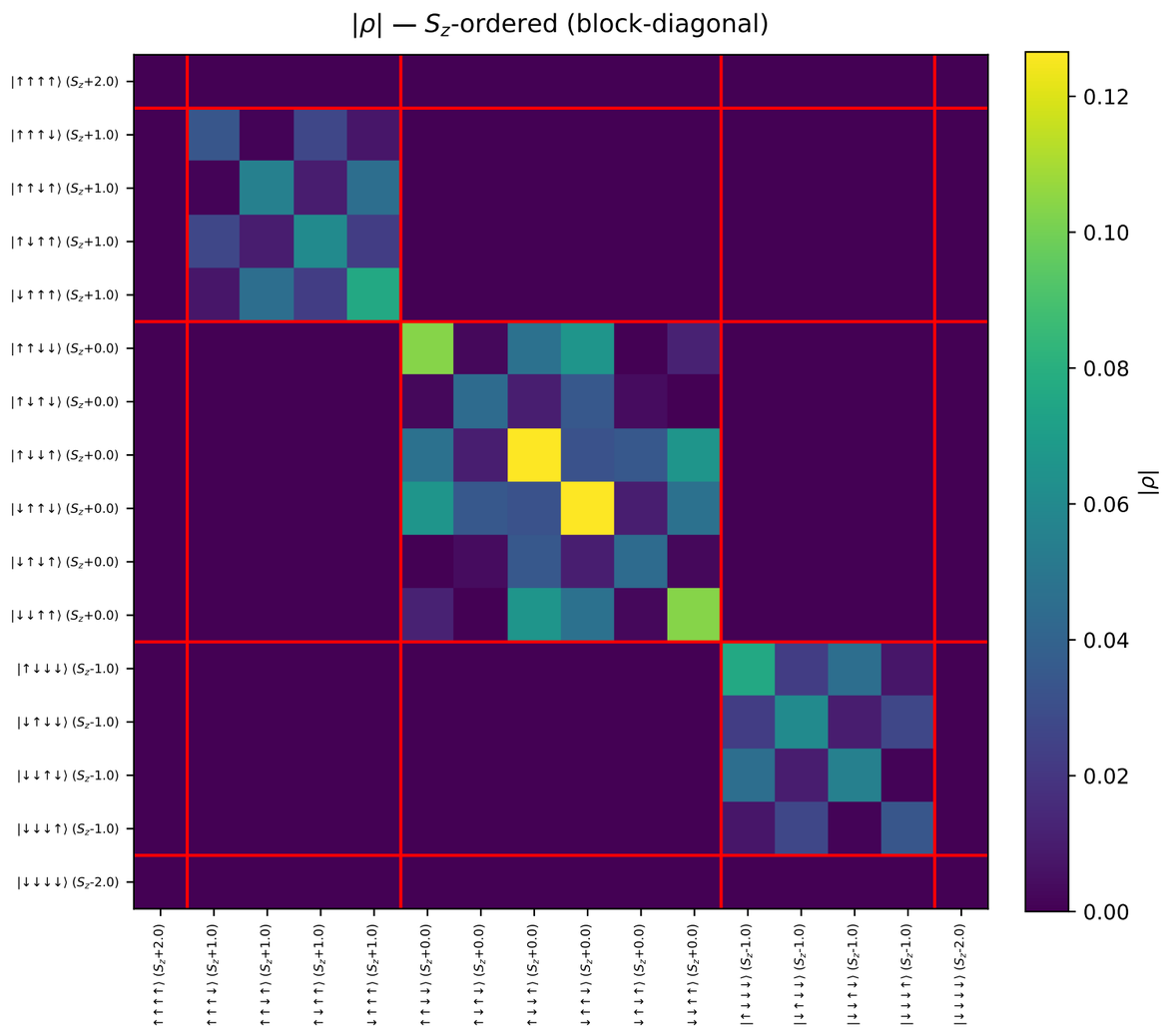


Trace=1.000000 Purity=0.154783 Entropy=1.988207 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.007463 ≤ C_{UB} = 0.271753 status: inconclusive

4-Body RDM (Sites 1,2,4,5)

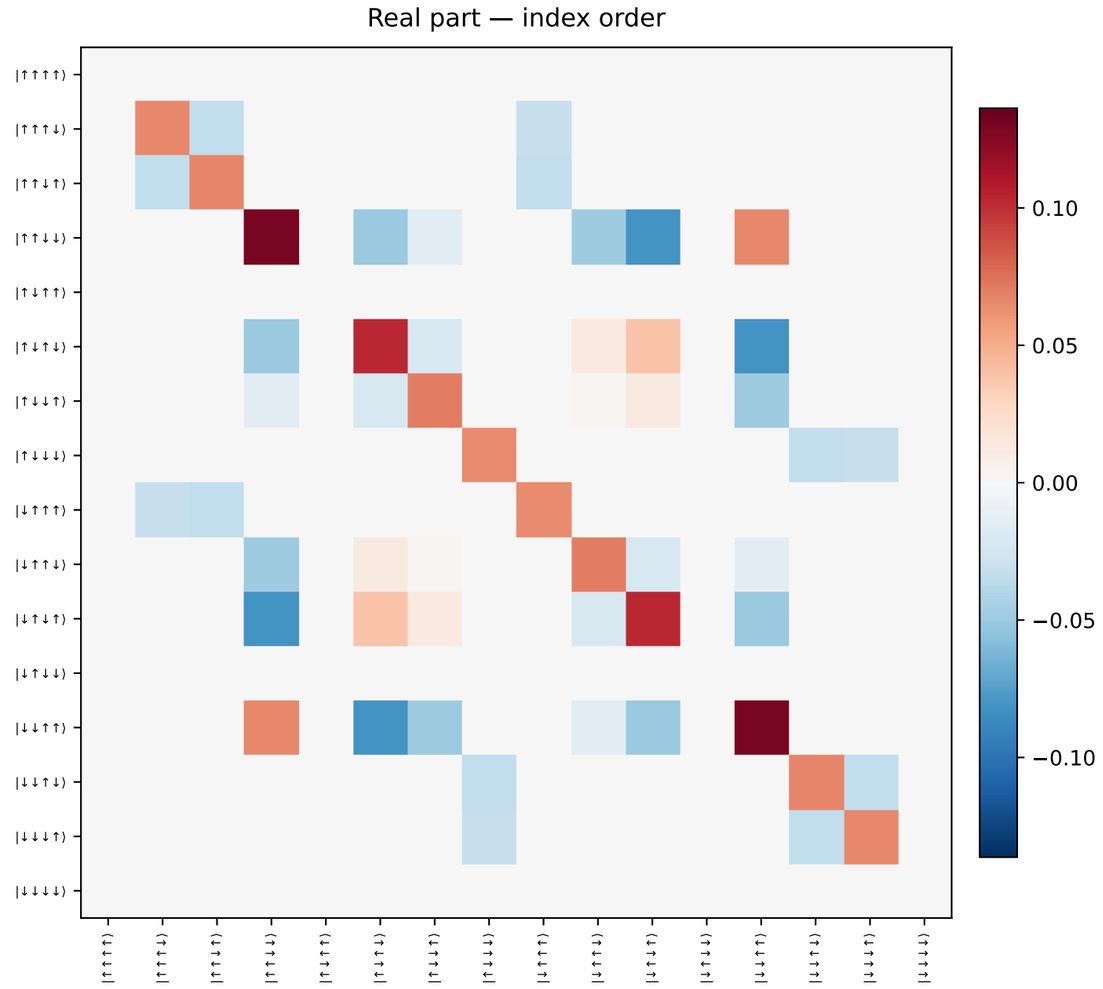


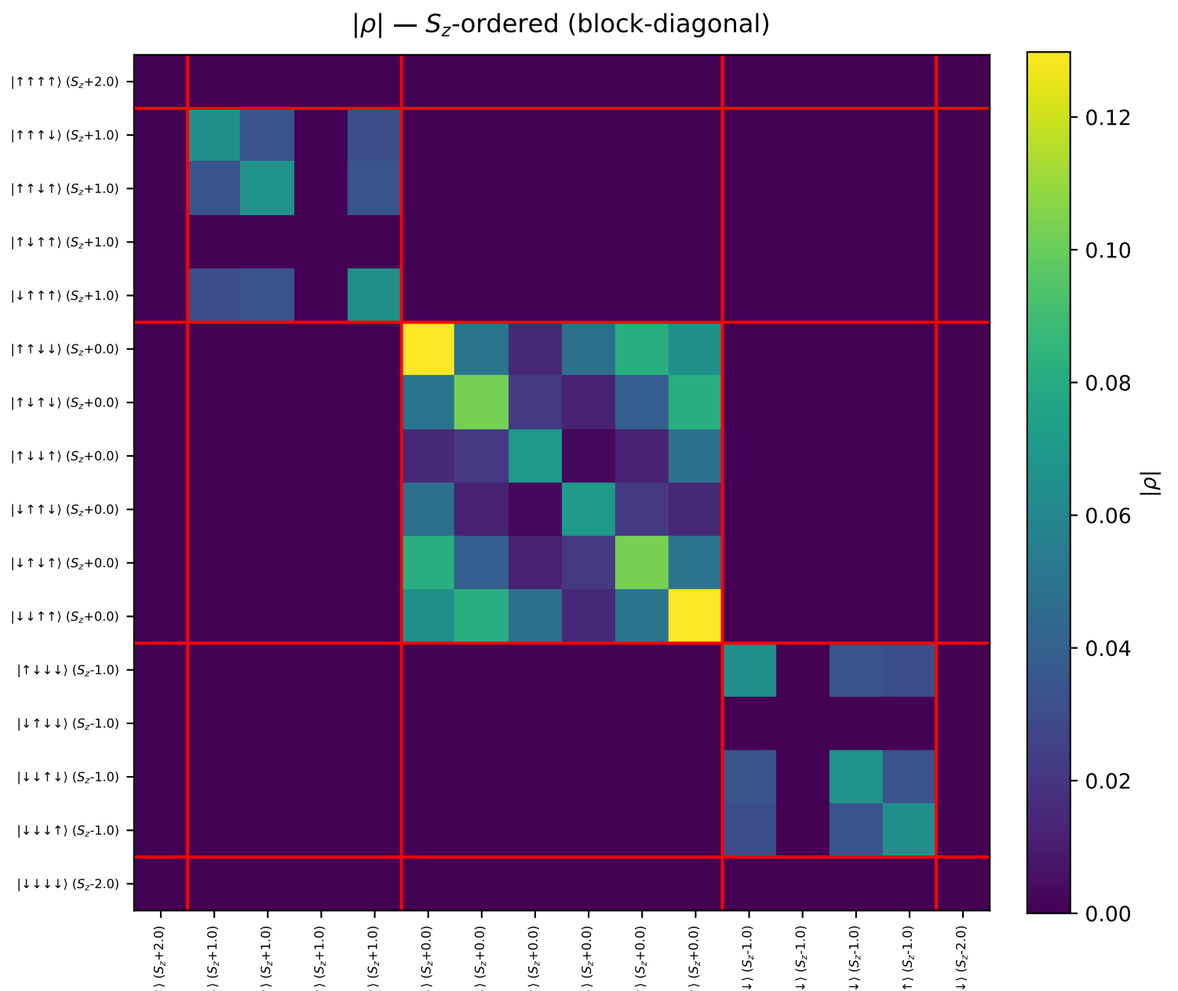


Trace=1.000000 Purity=0.132655 Entropy=2.197134 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.026273 ≤ C_{UB} = 0.315831 status: inconclusive

4-Body RDM (Sites 1,3,4,5)

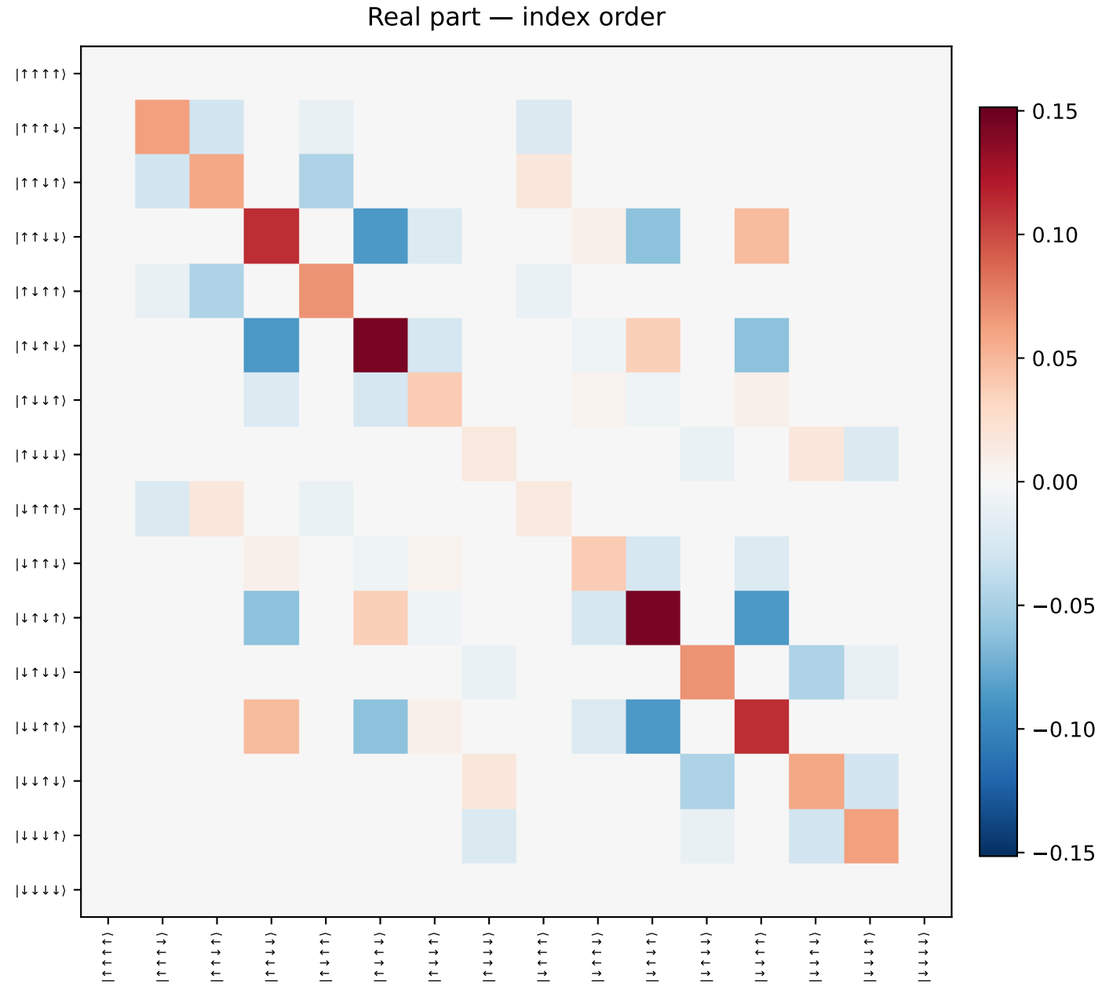


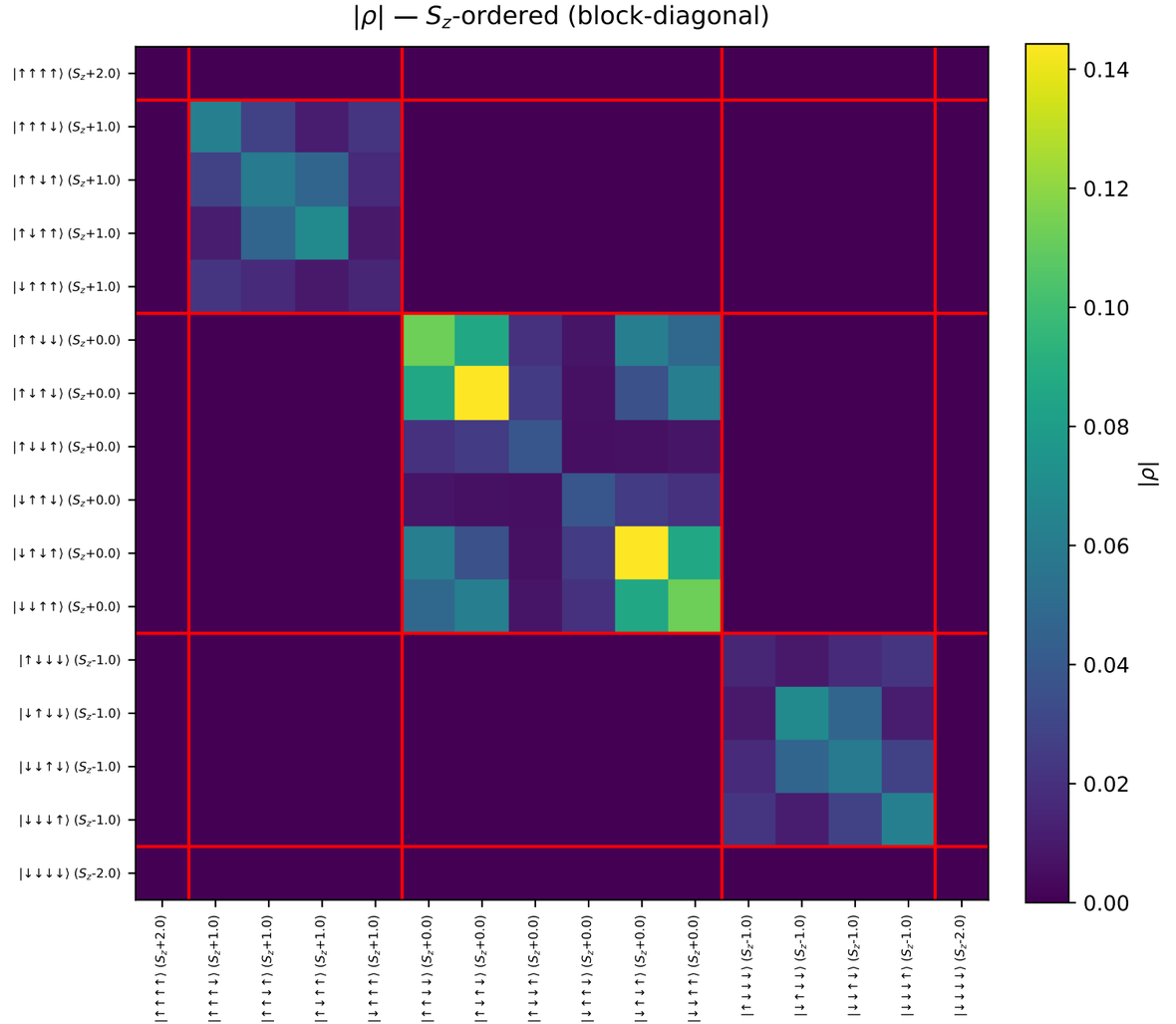


Trace=1.000000 Purity=0.164560 Entropy=1.961652 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.007917 ≤ C_{UB} = 0.287752 status: inconclusive

4-Body RDM (Sites 2,3,4,5)





Trace=1.000000 Purity=0.167789 Entropy=2.011575 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.048559 ≤ C_{UB} = 0.357304 status: inconclusive